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Databricks Fundamentals Bootcamp

Mohit Batra

Microsoft Certified Trainer

Day 2 of 2



Mohit Batra

- Founder, Crystal Talks
- Author
- Microsoft Certified Trainer
- Ex-Microsoft, Saxo Bank



linkedin.com/in/mohitbatra/



Learning Objectives

- Set up and navigate the Databricks platform
- Use Apache Spark to build ETL pipelines
- Reliably storing the data using Delta format
- Use Databricks Unity Catalog to manage and govern data
- Build production-ready pipelines using Databricks Workflows



Course Schedule – Day 1

Exploring, Cleaning, and Transforming Data with Databricks

- Introduction to Databricks
- Understanding Apache Spark architecture
- Setting up and exploring Databricks
- Reading data from multiple file formats
- Cleaning and transforming data using PySpark
- Working with Spark SQL and visualizing data



Course Schedule – Day 2

Storing Data with Delta Lake and Building Workflows with Databricks

- Introduction to Delta Lake
- Storing data in data lake using Delta format
- Working with Delta Lake features
- Working with Unity Catalog
- Building workflows with Databricks



Prerequisites

- Azure Subscription for setting up Azure Databricks
OR Databricks Free Edition

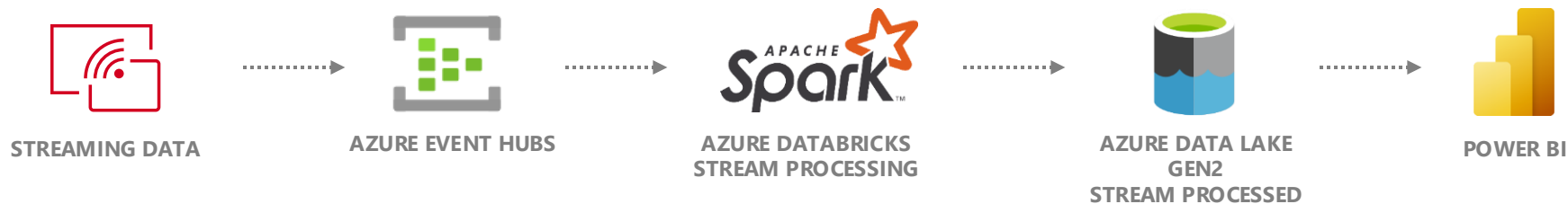




Introduction to Delta Lake

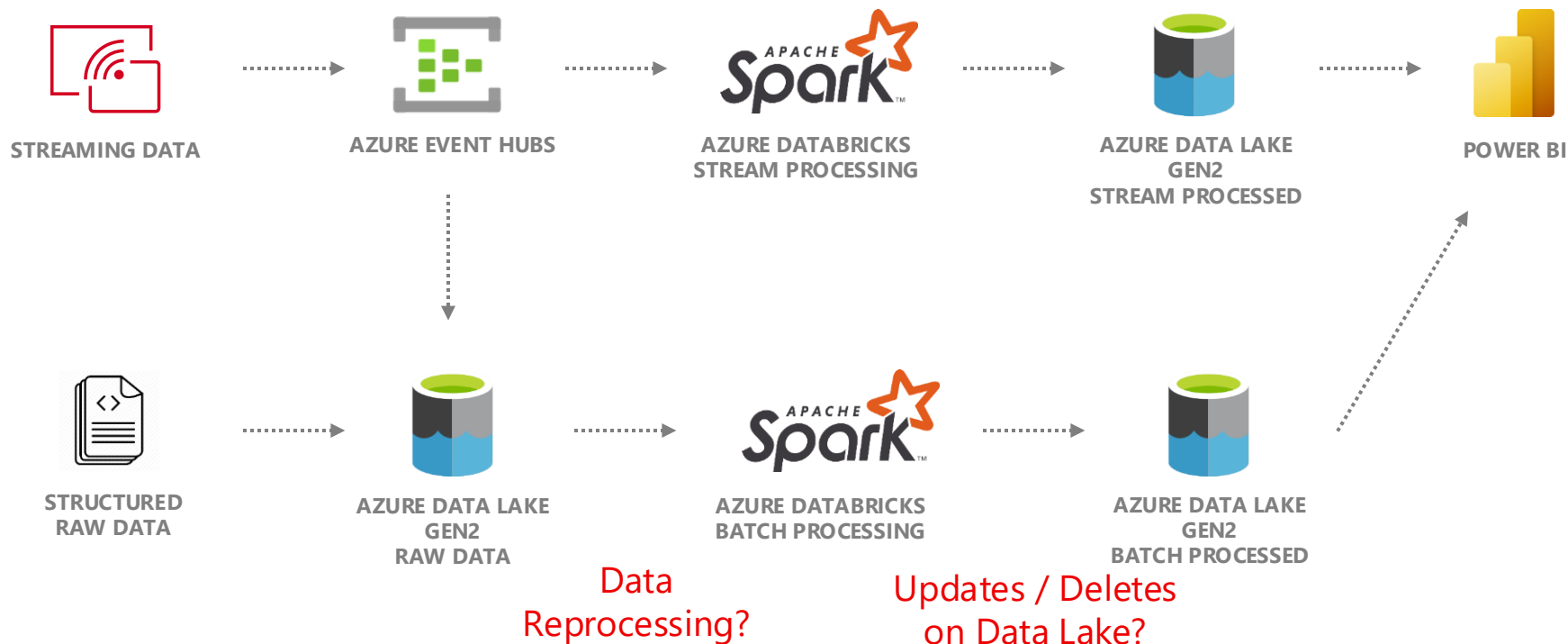


Using Data Lake for Storage





Using Data Lake for Storage





Challenges – Failure in Writing



3	Sabrina	24200
---	---------	-------

file2.parquet

4	Andrew	19000
5	Neha	32000

file3.parquet

1	Mohit	20000
2	Ivan	27500

file1.parquet

Inconsistent Data

Data Lake folder



Challenges – Parallel Reads/Writes



Reader

**Sees only 2 files –
Inconsistent Data**

3	Sabrina	24200
---	---------	-------

file2.parquet



Writer

4	Andrew	19000
5	Neha	32000

file3.parquet

1	Mohit	20000
2	Ivan	27500

file1.parquet

Data Lake folder



Challenges with Data Lake

Data Lake does not provide ACID guarantees

- Data Reliability
 - Data corruption because of failures – no rollback!
 - No data validation
 - Data read may not be consistent
- Updates / Deletes / Merge on files
- No constraints
 - No 'Not null' & 'Check' constraints
- Query Performance
 - Reading limited data is challenging
 - Lot of small files

Databricks



Data Engineering

AI and ML

Data Warehousing

OLTP Databases



Compute Options

Create clusters or use serverless compute

Photon
Engine



Unity Catalog

Unified security, cataloguing and governance



Delta Lake

Storage layer to bring reliability to Data Lakes



AI Integration



Monitoring



Security



Microsoft Azure



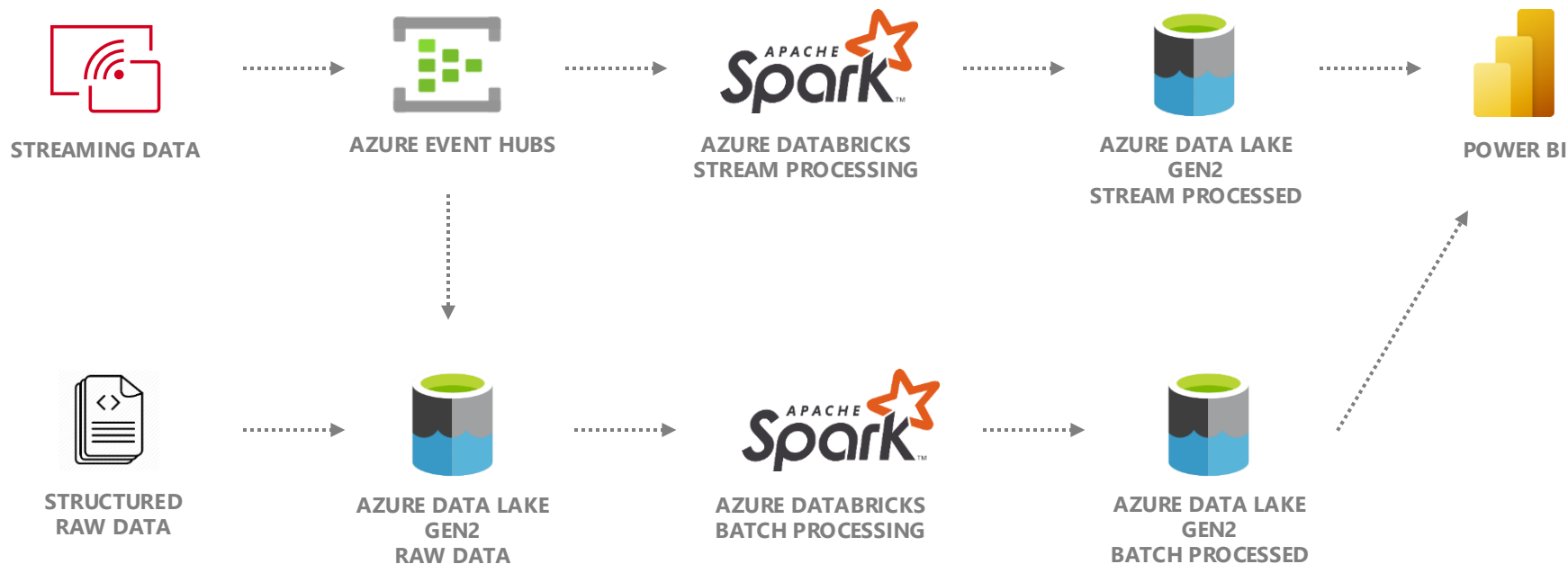
Amazon Web Services



Google Cloud

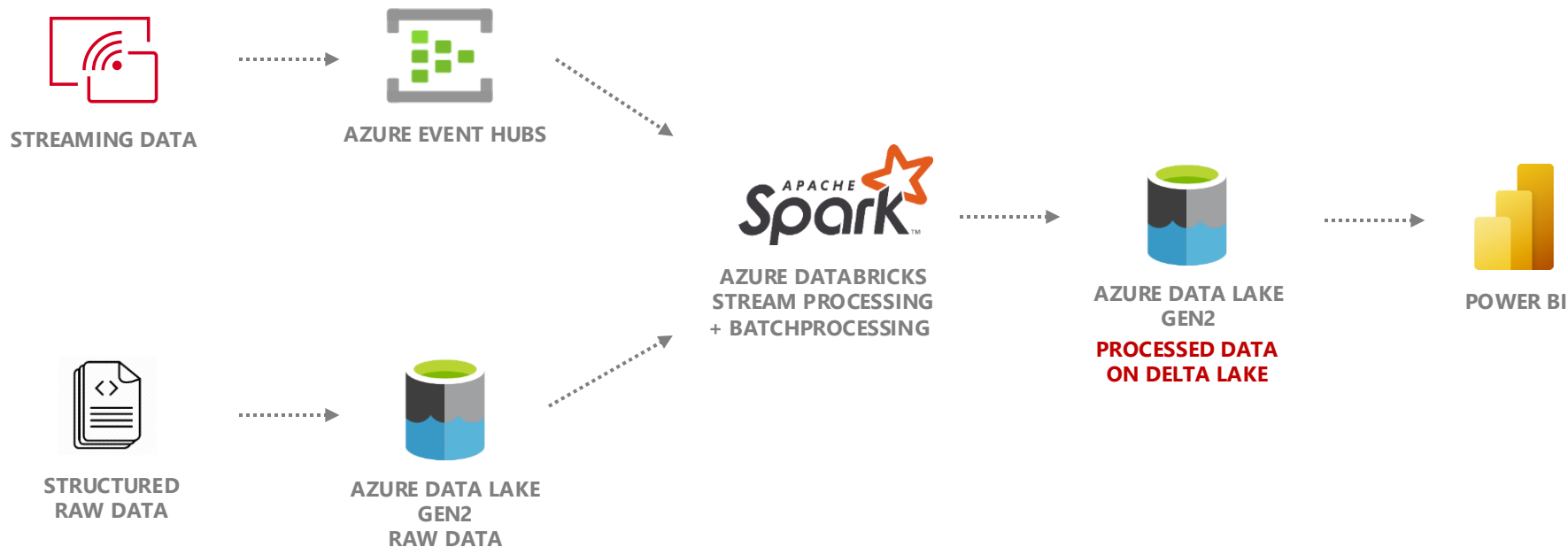


Using Data Lake for Storage





Using Data Lake for Storage



Delta Lake



User

1	Mohit	20000
2	Ivan	27500

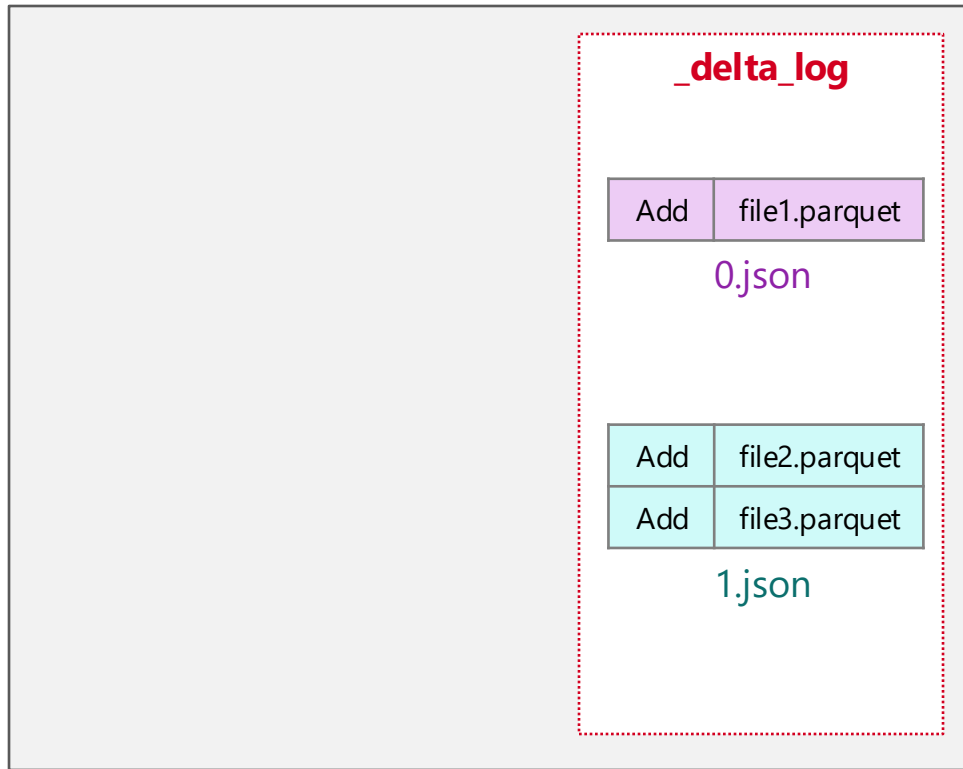
file1.parquet

3	Sabrina	24200
---	---------	-------

file2.parquet

4	Andrew	19000
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file3.parquet



Data Lake



Delta Lake – Failure in Writing



User

3	Sabrina	24200
---	---------	-------

file2.parquet

4	Andrew	19000
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file3.parquet

1	Mohit	20000
2	Ivan	27500

file1.parquet

_delta_log

Add	file1.parquet
-----	---------------

0.json

**No entry for
file2.parquet in
transaction log –
Consistent Data**

Data Lake



Delta Lake – Parallel Reads/Writes



Reader

**Since transaction log
only has 0.json, it
will only read file1 –
Consistent Data**

3	Sabrina	24200
---	---------	-------

file2.parquet



User

4	Andrew	19000
5	Neha	32000

file3.parquet

1	Mohit	20000
2	Ivan	27500

file1.parquet

_delta_log

Add	file1.parquet
-----	---------------

0.json

Data Lake

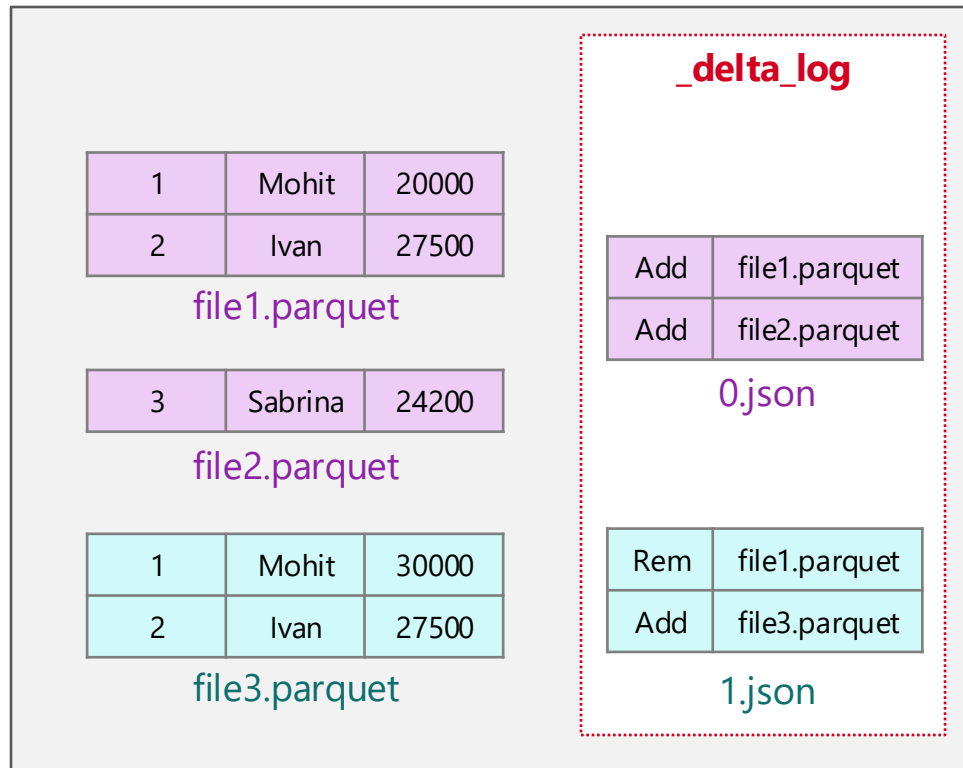


Delta Lake – Updating Record



User

**Update Amount to
30K where Id = 1**



Data Lake

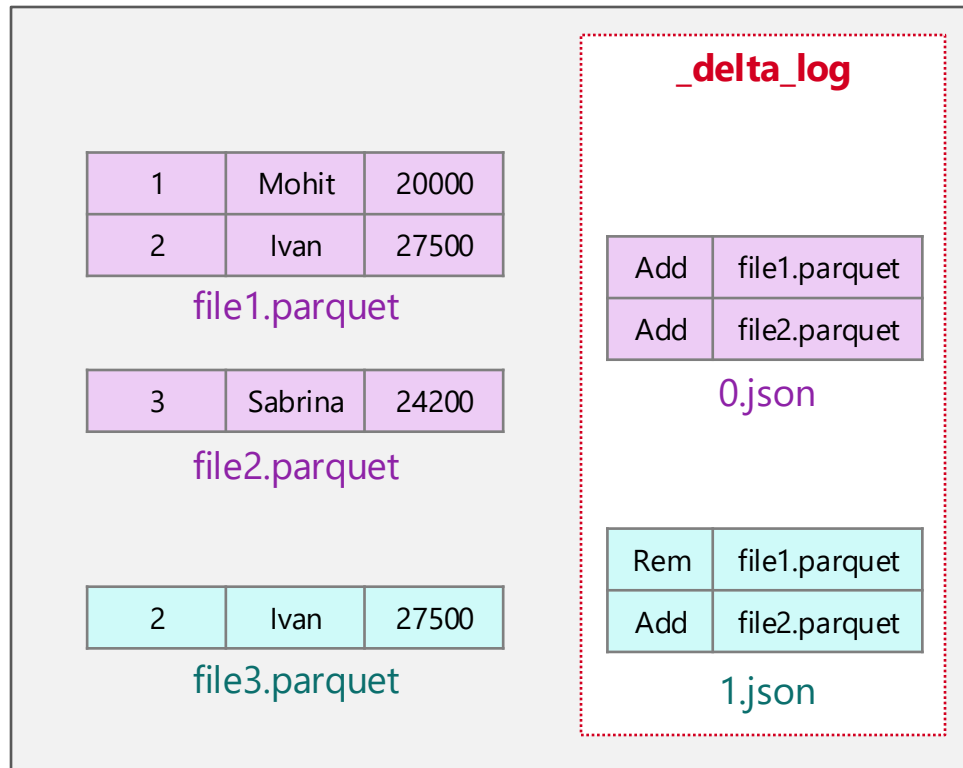


Delta Lake – Deleting Record



User

Delete where Id = 1



Data Lake



Delta Lake – Features

- Unified source & sink for Batch & Streaming Pipelines
- ACID transactions
- Multiple data pipelines can read and write at the same time
- Updates, deletes, merge
- Schema enforcement / evolution
- Audit history
- Time travel
- Versioning of data, enables rollback!
- Define column constraints



Question: Which of the following statements describes Delta Lake?

- Delta Lake is an open-source analytics engine used for big data workloads
- Delta Lake is an open format storage layer that delivers reliability, security, and performance
- Delta Lake is an open-source platform to help manage the complete machine learning lifecycle
- Delta Lake is an open format storage layer that processes data





Storing Data in Data Lake using Delta Format



Exercise

- Write DataFrames in Delta format
 - Create and managing Delta tables
 - Check audit history
-
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
 - Data Files @ <https://tinyurl.com/oreilly-databricks-bootcamp>



Question: Which of the following describes the storage organization of a Delta table?

- Delta tables are stored in a single file that contains data, history, metadata, and other attributes
- Delta tables are stored in a single file that contains only the data stored within the table
- Delta tables are stored in a collection of files that contain data, history, metadata, and other attributes
- Delta tables are stored in a collection of files that contain only the data stored within the table





Question: Delta Lake uses which underlying file format for storing data?

- JSON
- Parquet
- Avro
- CSV





Working with Delta Lake Features



Question: Which feature of Delta Lake allows querying older versions of data?

- Schema Enforcement
- Time Travel
- Delta Snapshots
- Restore





Question: What happens if schema enforcement is enabled and data does not match the schema?

- Delta Lake automatically adjusts the schema
- The mismatched data is ignored
- The whole write operation is rejected
- The entire table is dropped





Exercise

- Perform CRUD operations
 - Apply schema enforcement and evolution
 - Perform time travel
 - Set up table constraints
-
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
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Working with Unity Catalog

Databricks



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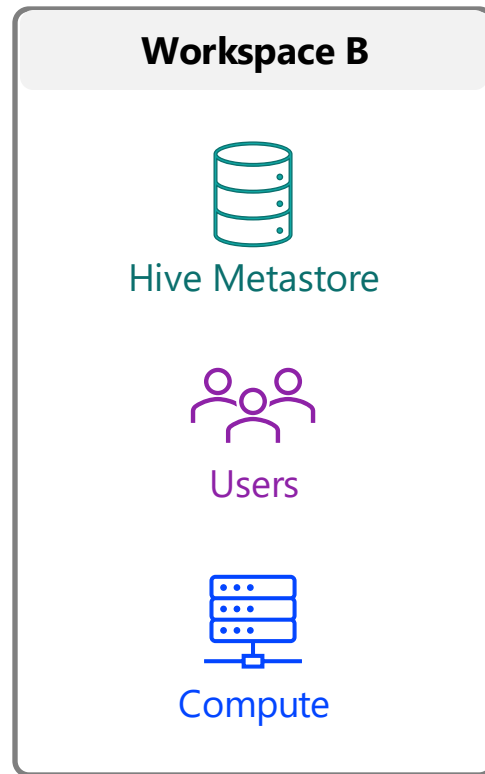
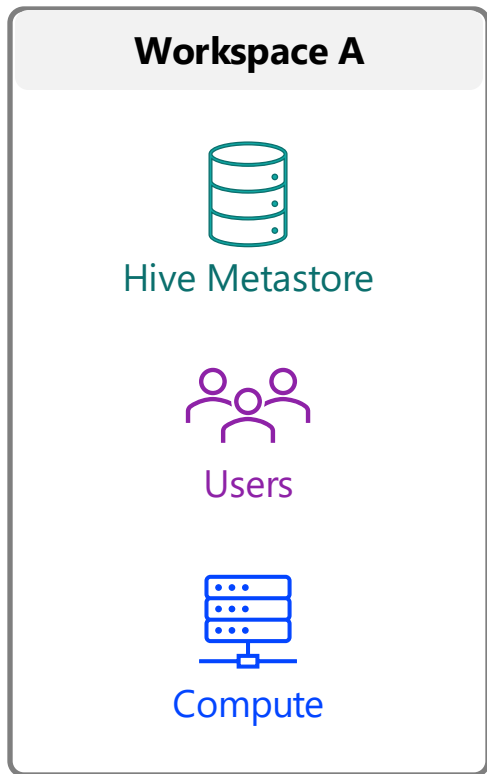


Amazon Web Services

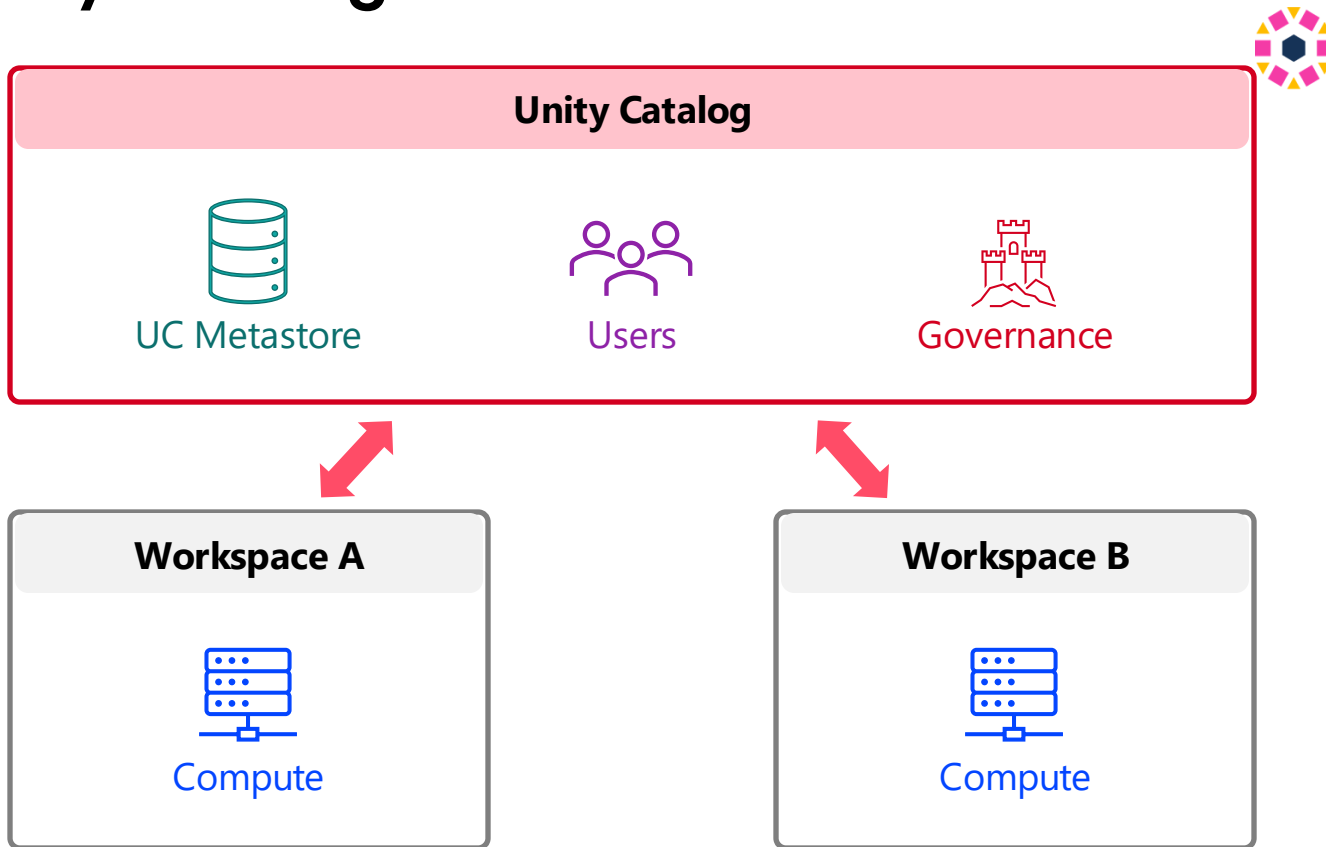


Google Cloud

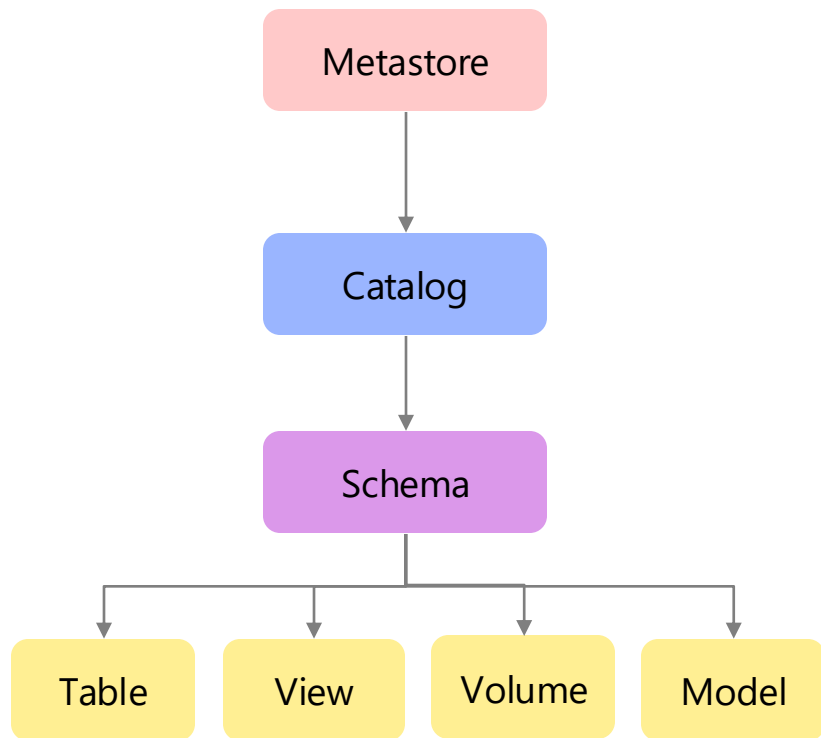
Without Unity Catalog



With Unity Catalog

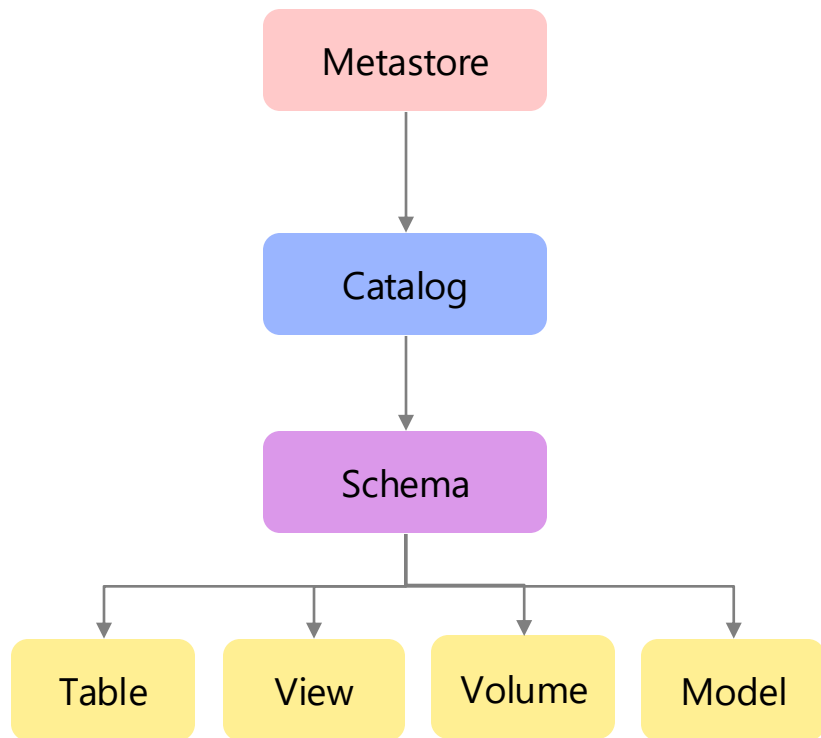


Unity Catalog Object Model



- One per region / Root container to store metadata
- Workspaces can connect to metastore in the same region only
- Logical unit of data isolation (like org units)
- Catalogs in same metastore can be shared between workspaces
- Legacy Hive_metastore is a catalog, not a metastore
- Also known as databases
- Create logical categories (like projects)
- Create and register assets

Unity Catalog Object Model



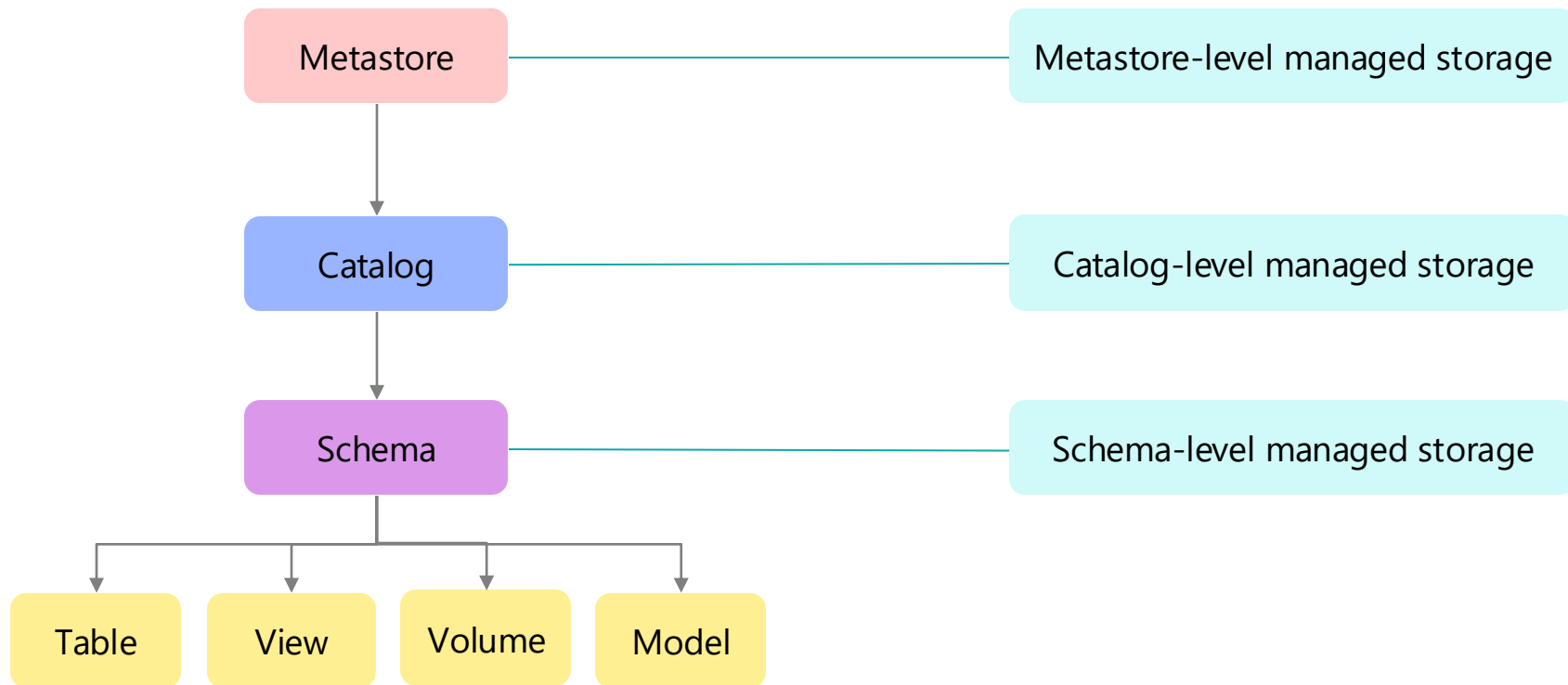
Within a workspace that is connected to Metastore, there is a three-level namespace

SELECT * FROM `catalog.schema.table`

Example - `finance.retail.customers`

Unity Catalog Storage

Cloud Storage
(ADLS, S3, GCS)





Unity Catalog Features

- **Security**
 - Define data access policies
 - Manage permissions for identities
 - Manage connections to external storage locations
- **Built-in Auditing and Lineage**
 - Captures audit logs that records access to data by users
 - Captures lineage that tracks how data is moving between assets
- **Data Discovery**
 - Allows you to tag assets with additional information (like PII)
 - Business users can search for assets
- **System Tables**
 - Allows you to query logs, billing, lineage, etc.





Question: Which of the following is NOT a feature of Databricks Unity Catalog?

- Centralized governance and access control
- Fine-grained permissions at the table and column level
- Built-in dashboard creation for visualization
- Data lineage tracking





Exercise

- Create catalog and schema
- Add a table to catalog
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
- Data Files @ <https://tinyurl.com/oreilly-databricks-bootcamp>



Building Workflows with Databricks

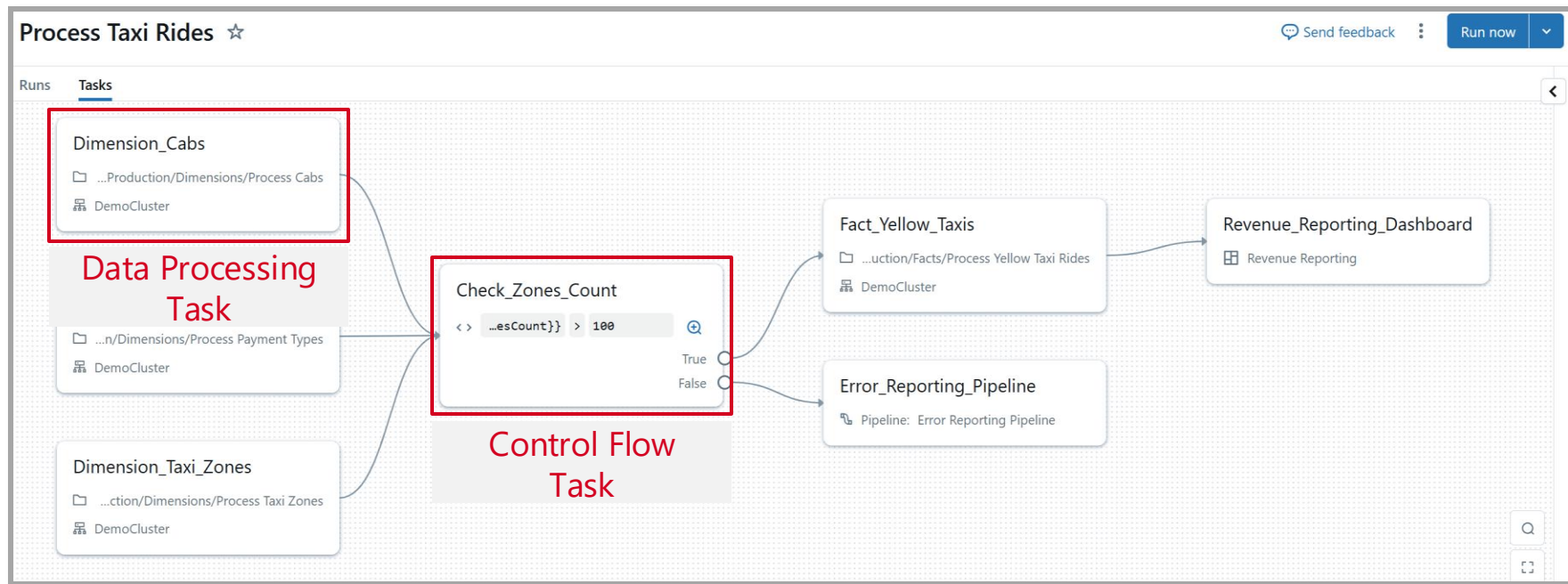


Workflow Types

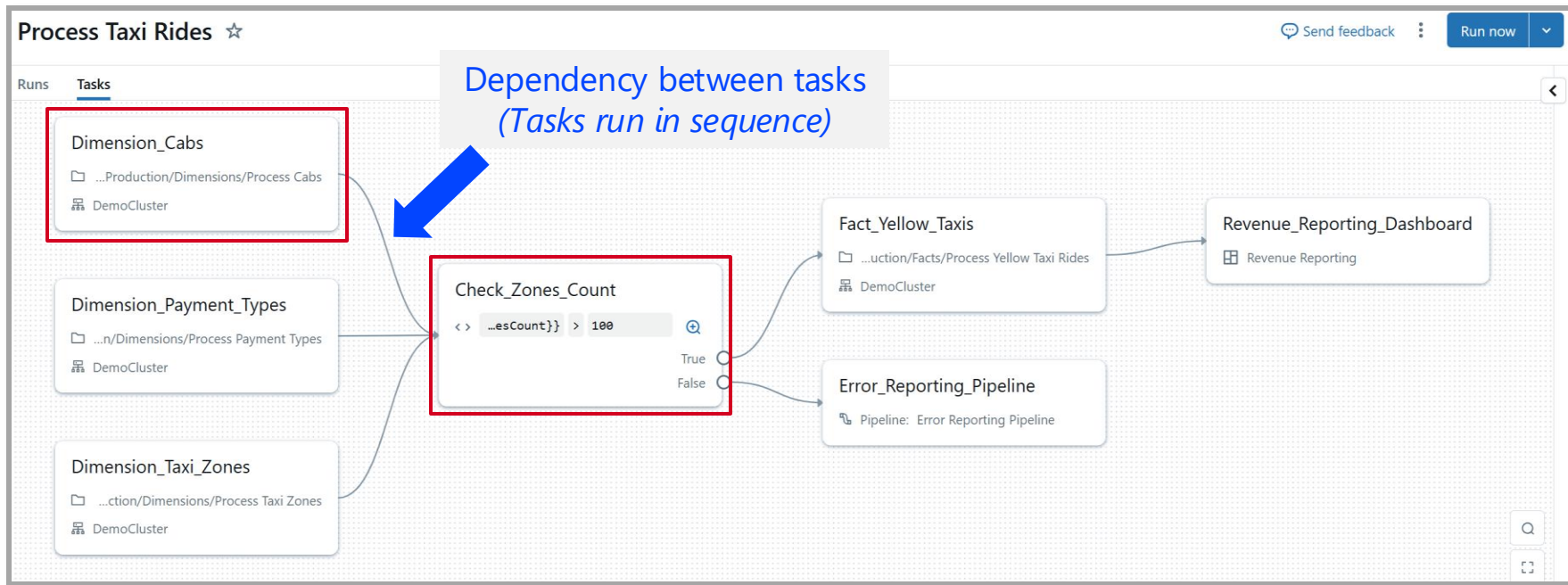
- **Ingestion Pipeline (Lakeflow Connect)**
 - Ingest data from variety of data sources
 - Connectors from Databricks and Fivetran are available
- **ETL Pipeline (Lakeflow Declarative Pipelines)**
 - ETL framework to build declarative data pipelines
 - Use it for performing transformations, applying data quality checks, managing dependencies, etc.
- **Job (Lakeflow Jobs)**
 - Orchestrate notebooks, pipelines, queries, and more

Used to build and automate workflows to perform data processing tasks.

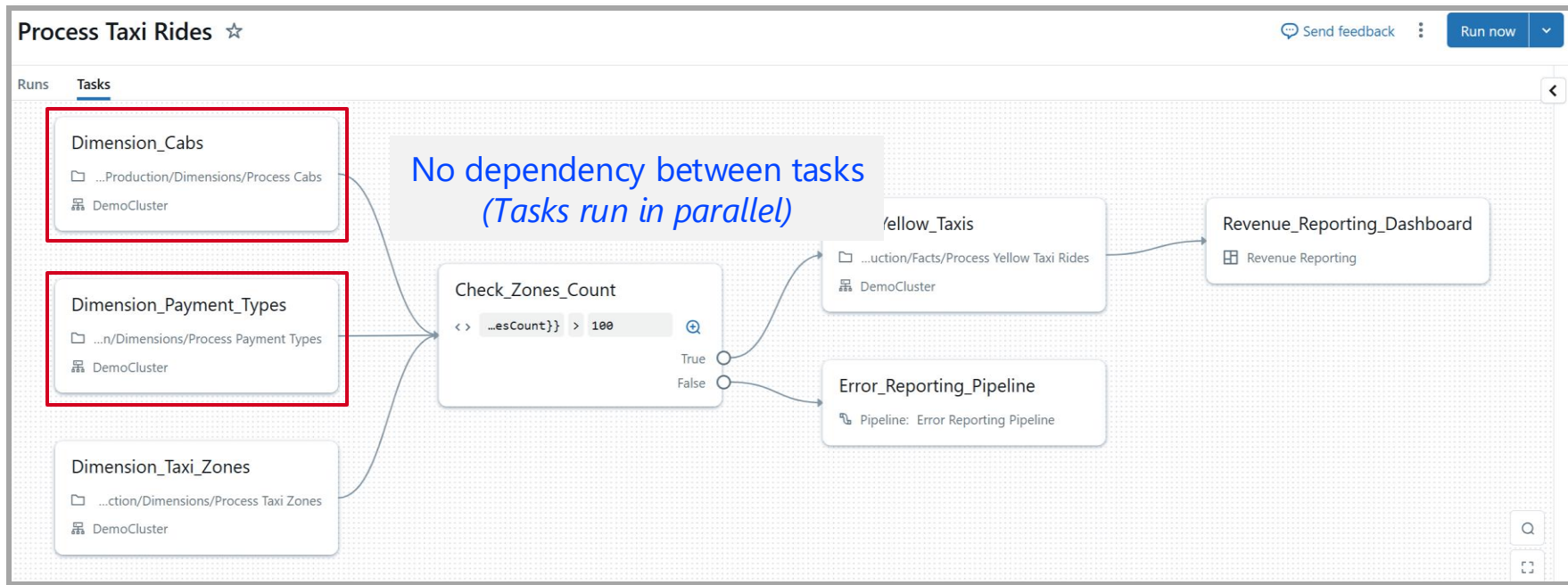
Sample Job



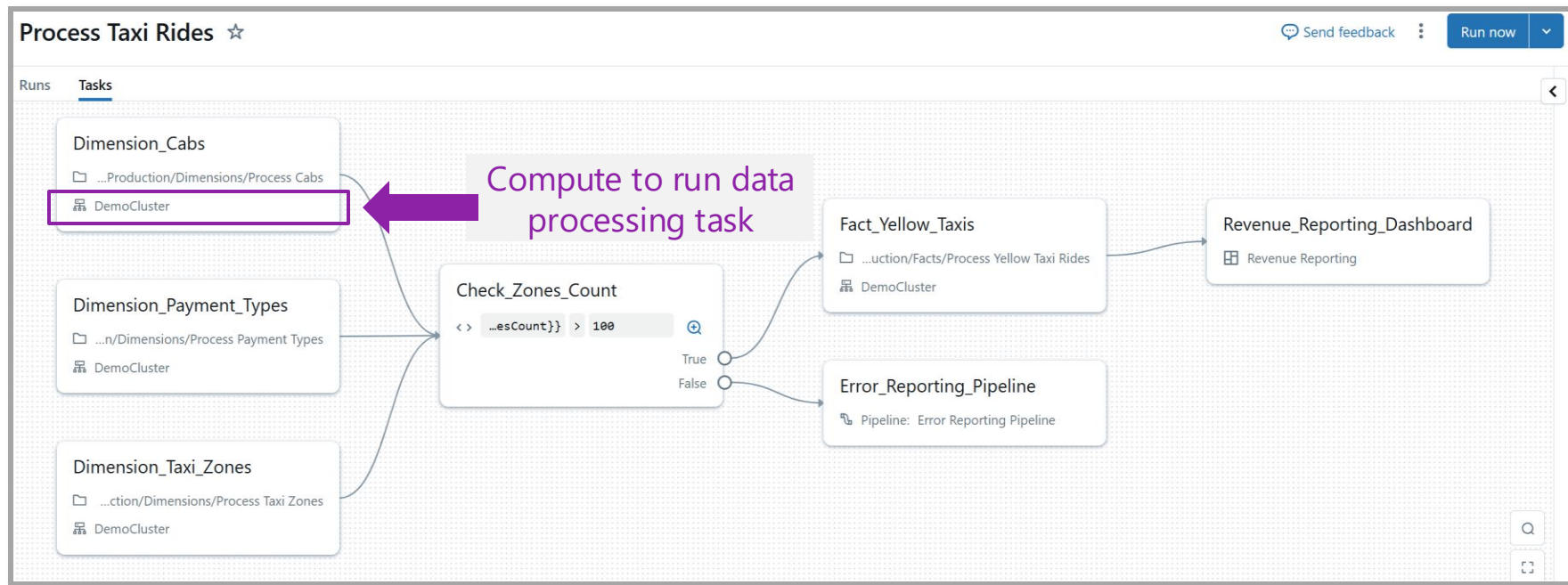
Sample Job



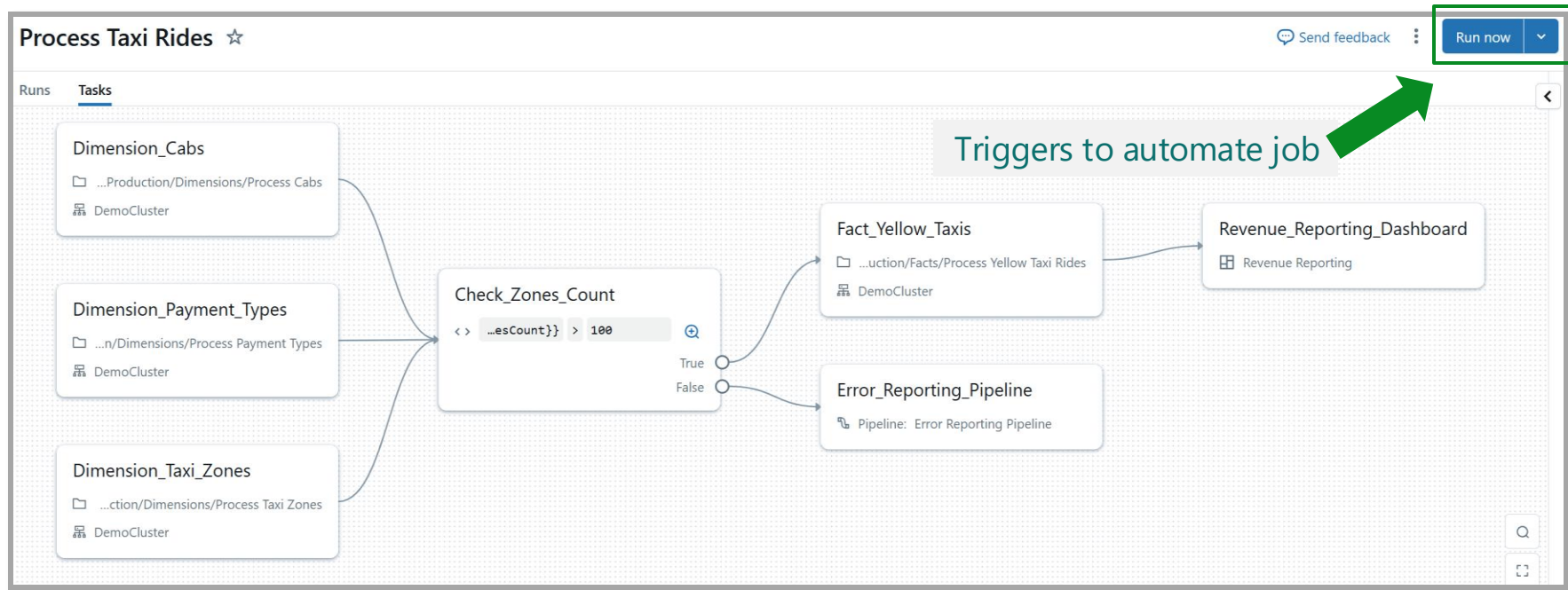
Sample Job



Sample Job



Sample Job





Question: A data engineer has set up two Jobs that each run nightly. The first Job starts at 12:00 AM, and it usually completes in about 20 minutes. The second Job depends on the first Job, and it starts at 12:30 AM. Sometimes, the second Job fails when the first Job does not complete by 12:30 AM.

Which approach can the data engineer use to avoid this problem?

- They can utilize multiple tasks in a single job with a linear dependency
- They can use job cluster to help the Jobs run more efficiently
- They can set up a retry policy on the first Job to help it run more quickly
- They can limit the size of the output in the second Job so that it will not fail as easily





Exercise

- Create a job by adding multiple tasks
 - Create a job cluster
 - Schedule job
-
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
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Key Takeaways – Day 2

Storing Data with Delta Lake and Building Workflows with Databricks

- Introduction to Delta Lake
- Stored data in data lake using Delta format
- Worked with Delta Lake features
- Worked with Unity Catalog
- Built workflows with Databricks



Mohit Batra

Feel free to reach out...



[linkedin.com/in/mohitbatra/](https://www.linkedin.com/in/mohitbatra/)



The image features the O'Reilly logo in white text on a blue background. The background has a gradient from dark blue on the left to light blue on the right. There are two large, faint, semi-transparent blue circles on the left side of the image. The text "O'REILLY" is in a bold, sans-serif font, with a registered trademark symbol (®) at the end.

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